

Pre-A Q3LOCK Zero-Source Fixed-Order Fourth-Order Linkedness and Orbit-Locality Boundary Checkpoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

first issued 2026-08-12 · this version issued 2026-08-12 · v1.5

1 Purpose and scope

This R-167-only gate-level checkpoint synthesizes EXP-000826 as corrected by EXP-000827 and records reusable result R-167 v2.6 without changing a claim card or tier. The result remains additive, T0, and `claim_bearing: false`. Its stable result ID is

[PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#).

Append-only provenance is retained. Changelog event 617 and EXP-000826 remain immutable history. Event 618 and EXP-000827 supersede the earlier wording in exactly two respects: the first two Bravyi-DiVincenzo-Loss children are zero-source fixed- M Q3 statements, and the bare-edge third/fourth QPS values are replaced by the augmented-edge values derived below. The correction creates no new result number or version. Event 617 must be read only as corrected by event 618.

R-167 v2.6 has four distinct positive scopes. In the **zero-source fixed-Ritz** setting, **each separately fixed** standard-SW coefficient has connected support. At fixed (n, M) and sufficiently small coupling, fixed-order effective ground energies have a volume-extensive error estimate. For the actual zero-source Q3 blocks, only orders three and four have the displayed volume- and Ritz-uniform coefficient-QPS bounds and Ritz removal. At fixed finite Λ, M , the complete sequential fourth-order coefficient and its standard-SW gauge crosswalk are exact finite-matrix statements.

None of these scopes takes n to infinity, proves convergence of an all-order series, verifies physical $\lambda=1$ smallness, transfers a spin phase, or proves an oscillator GNS gap. R-168 v1.3 remains historical and is not reissued. No R-168 count, hash, theorem, or lifecycle field belongs here.

2 Exactly four CLOSED children, two negatives, and five OPEN parents

The four children closed in their corrected scopes are exactly

1. [PA-CP1-ST8-Q3LOCK-FIXED-RITZ-STANDARD-SW-EVERY-FIXED-ORDER-LINKED-CLUSTER-COEFFICIENTS](#);
2. [PA-CP1-ST8-Q3LOCK-FIXED-RITZ-EACH-FIXED-ORDER-SMALL-COUPPING-VOLUME-EXTENSIVE-SW-GROUND-ENERGY-APPROXIMATION](#);
3. [PA-CP1-ST8-Q3LOCK-ZERO-SOURCE-Q3-THIRD-FOURTH-ORDER-COEFFICIENT-QPS-AND-RITZ-CUTOFF](#);

4. [PA-CP1-ST8-Q3LOCK-FIXED-FINITE-VOLUME-AND-RITZ-COMPLETE-FOURTH-ORDER-SEQUENTIAL-LOW-BLOCK-AND-STANDARD-SW-GAUGE-CROSSWALK.](#)

The first two IDs carry the explicit zero-source fixed- M qualifier from EXP-000827. The third is actual Q3 but only through orders three and four. The fourth is fixed finite Λ, M algebra. The two negative authorities are exactly

1. [NG-2026-08-12-PRE-A-ST8-Q3LOCK-LOW-HIGH-RITZ-TAIL-AUTOMATIC-UNIFORM-HIGH-HIGH-INSERTION-CUTOFF;](#)
2. [NG-2026-08-12-PRE-A-ST8-Q3LOCK-ORBIT-SMEAR-SEED-SUPPORT-AUTOMATIC-SPATIAL-LOCAL-NET.](#)

They refute only the named automatic implications. They do not invalidate the weighted Q3 fixed-order route or rule out an orbit-smear carrier equipped with an independent propagation theorem. The five load-bearing gates recorded with this checkpoint remain **OPEN**:

1. [PA-CP1-ST8-Q3LOCK-CONNECTED-RANK-TWO-OSCILLATOR-ELIMINATION-QPS-NORM-AND-CUTOFF-COMPATIBILITY;](#)
2. [PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;](#)
3. [PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;](#)
4. [PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;](#)
5. [PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE.](#)

Neither the four scoped closures nor the two implication failures close any parent in this list.

3 Exact zero-source augmented-edge mapping

Fix one parity-preserving onsite Ritz cutoff M on a finite connected zero-source Q3 graph $G = (\Lambda, E)$, with $\deg(x) \geq 1$ and maximum degree z . Put

$$H_0 = \sum_x k_{x,M} \geq 0, \quad P_0 = \bigotimes_x P_x, \quad k_{x,M} \geq \Gamma(1 - P_x). \quad (3.1)$$

Each P_x has rank two, which is allowed by the finite-rank onsite ground projector formulation. Choose nonnegative incident weights satisfying

$$\omega_{x,e} \geq 0, \quad \sum_{e \ni x} \omega_{x,e} = 1. \quad (3.2)$$

On a periodic coordination- z graph use $\omega_{x,e} = 1/z$. For $e = \{x, y\}$, define

$$\tilde{B}_{e,M} = B_{e,M} + \delta_1(\omega_{x,e} P_{1,x} + \omega_{y,e} P_{1,y}), \quad (3.3)$$

$$V_M = \sum_e \tilde{B}_{e,M}, \quad J_M = \max_x \sum_{e \ni x} \|\tilde{B}_{e,M}\|, \quad (3.4)$$

$$H_M(\lambda) = \sum_x k_{x,M} + \lambda V_M. \quad (3.5)$$

The incident partition gives the exact allocation identity

$$\sum_{e=\{x,y\}} \delta_1(\omega_{x,e}P_{1,x} + \omega_{y,e}P_{1,y}) = \delta_1 \sum_x P_{1,x}. \quad (3.6)$$

Consequently $H_M(1)$ is the physical zero-source fixed- M Q3 Hamiltonian, up to the scalar already removed from every onsite term. The auxiliary $k = h_{\text{site}} - \epsilon_0 - \delta_1 P_1$ has the declared rank-two zero eigenspace; (3.6) restores the physical splitting at $\lambda = 1$. Omitting this allocation was the mapping defect corrected by EXP-000827.

4 Fixed-order linked coefficients and energy accuracy

The literature authority is Sergey Bravyi, David P. DiVincenzo, and Daniel Loss, *Schrieffer–Wolff transformation for quantum many-body systems*, Annals of Physics **326** (2011), 2793–2826, <https://doi.org/10.1016/j.aop.2011.06.004>, arXiv:1105.0675. Section 4.3, especially Theorem 2 on arXiv PDF page 33, supplies the coefficientwise linked-cluster statement. The separately scoped fixed-order energy and standard/local gauge comparisons are those cited in the proof certificate.

At each fixed M , the zero-source augmented family (3.3)–(3.5) satisfies the finite-dimensional hypotheses. For every separately fixed order n , the standard-SW multivariate coefficient decomposes as

$$H_{\text{eff},n} = \sum_C K_{n,C}, \quad K_{n,C} = 0 \quad \text{unless } C \text{ is connected and } |C| \leq n. \quad (4.1)$$

The operator $K_{n,C}$ acts only on the vertices of C . A nonzero coefficient has support at most $n + 1$ and diameter at most n . A safe, nonoptimal rooted ordered count is

$$N_{n,\text{root}} \leq (n!)^2 z^n. \quad (4.2)$$

At fourth order this is $576z^4$; paying support at most five gives the safe $2880z^4$ QPS count.

Normalize the fixed- M augmented interaction to local strength one and put $\eta = \lambda J_M$. For every separately fixed (n, M) , there are positive constants $\epsilon_{c,n,M}$ and $C_{n,M}$, independent of $|\Lambda|$, such that $|\eta| < \epsilon_{c,n,M}$ implies

$$\begin{aligned} |E_0(H_M(\lambda)) - E_0(H_{\text{eff,std}}^{[n]}(\lambda))| &\leq C_{n,M} |\Lambda| |\eta|^{n+1}, \\ |E_0(H_M(\lambda)) - E_0(H_{\text{eff,loc}}^{[n]}(\lambda))| &\leq C_{n,M} |\Lambda| |\eta|^{n+1}. \end{aligned} \quad (4.3)$$

The physical fixed- M point has $\lambda = 1$ and $\eta = J_M$, but no proof here places J_M below the required threshold. Equation (4.1) is coefficientwise and proves no convergence of the full formal series. Equation (4.3) is separately fixed in n, M ; it is not an all-order volume-uniform convergence theorem.

5 Actual Q3 weighted blocks and seven corrected rationals

At zero source put $k_f = k_x + k_y$, $P_f = P_x P_y$, and $Q_f = 1 - P_f$. For the bare and augmented edges define

$$\begin{aligned} A_f &= P_f B_f P_f, & \tilde{A}_f &= P_f \tilde{B}_f P_f, \\ L_f &= Q_f B_f P_f = Q_f \tilde{B}_f P_f, & U_f &= Q_f B_f Q_f, \quad \tilde{U}_f = Q_f \tilde{B}_f Q_f. \end{aligned} \quad (5.1)$$

The allocated onsite term is block diagonal in the local P_f/Q_f decomposition, so the cross leg is unchanged. Since $P_{1,x}P_{1,y} \leq P_f$, the eigenvalue-two branch of $P_{1,x} + P_{1,y}$ lies in P_f and is removed on Q_f . Hence

$$0 \leq Q_f(P_{1,x} + P_{1,y})Q_f \leq Q_f \leq \frac{k_f}{\Gamma}, \quad \|P_f(P_{1,x} + P_{1,y})P_f\| = 2. \quad (5.2)$$

Starting from

$$\|A_f\| \leq a, \quad \|L_f\| \leq \epsilon, \quad 0 \leq U_f \leq \rho_b k_f, \quad (5.3)$$

the periodic choice gives

$$\tilde{a} = a + \frac{2\delta_1}{z}, \quad \tilde{\rho} = \rho_b + \frac{\delta_1}{z\Gamma}. \quad (5.4)$$

With $K \geq \Gamma Q$, $R = (QKQ)^{-1}$, and $W_f = R^{1/2}Q\tilde{B}_fQR^{1/2}$, the augmented local-high and outside-high locally-low sectors are orthogonal local diagonal blocks. Their contribution is a maximum, not a sum. Adding the two unchanged cross legs gives

$$\|W_f\| \leq w_{\text{tilde}} := \max\left(\tilde{\rho}, \frac{\tilde{a}}{\Gamma}\right) + \frac{2\epsilon}{\Gamma}. \quad (5.5)$$

For

$$\rho_b = \frac{15201}{156250}, \quad a = \frac{96139}{1500000}, \quad \delta_1 = \frac{1}{10000}, \quad \epsilon = \frac{23}{6250}, \quad \Gamma = 100, \quad z = 6, \quad e^{a_{\text{QPS}}} = 2, \quad (5.6)$$

the $\tilde{\rho}$ branch dominates $\tilde{a}/\Gamma$. The seven corrected exact rational oracles are

```
a_tilde = 32063/500000
rho_tilde = 2918597/30000000
w_tilde = 584161/6000000
per3 = 3888206603/292968750000000000
QPS3 = 1411418996889/381469726562500
per4 = 28578738599866921/21972656250000000000000000
QPS4 = 2314877826589220601/29802322387695312500
```

Thus event 617's third-order value 1420642437753/381469726562500 is superseded by 1411418996889/381469726562500; its fourth-order value 2345231533369183929/29802322387695312500 is superseded by 2314877826589220601/29802322387695312500.

Put $T_e = Q\tilde{B}_eP$ and $E_e = R^{1/2}T_e$, so $|E_e| \leq \epsilon/\sqrt{\Gamma}$. The third-order envelope and QPS bound are

$$p_3 = \epsilon^2 \left(\frac{w_{\text{tilde}}}{\Gamma} + \frac{\tilde{a}}{\Gamma^2} \right), \quad (5.7)$$

$$\|\Theta^{(3)}\|_{a_{\text{QPS}}} \leq 48z(2z-1)^2 e^{3a_{\text{QPS}}} p_3. \quad (5.8)$$

The exact factor is 278784, yielding QPS3 above. For fourth order use the actual augmented blocks $A = P\tilde{V}P$, $C = Q\tilde{V}Q$, with $\tilde{V} = \sum_f \tilde{B}_f$, and define

$$B = T^*RT, \quad F = T^*R^2T, \quad S = RTA - CRT. \quad (5.9)$$

Then

$$p_4 = \frac{\epsilon^4}{\Gamma^3} + \frac{\epsilon^2}{\Gamma} \left(w_{\text{tilde}} + \frac{\tilde{a}}{\Gamma} \right)^2, \quad (5.10)$$

$$\|\Theta^{(4)}\|_{a_{\text{QPS}}} \leq 2880z^4 e^{4a_{\text{QPS}}} p_4. \quad (5.11)$$

The exact factor is 59719680, yielding QPS4 above. Direct additivity and the triangle inequality over connected ordered contributions justify this sequential coefficient count. It is not based on the false claim that every individual disconnected ordering vanishes.

6 Complete fourth order and the standard-SW gauge crosswalk

At fixed finite Λ, M , use the actual augmented family of Section 3. Let $D = RT$, $G = D - D^*$, $Z = RS$, and $G_2 = Z - Z^*$. Expanding the two sequential rotations through order four gives

$$\Theta_{\text{seq}}^{(4)} = \frac{1}{8}P \text{ad}_G^3(V_{\text{od}})P + \frac{1}{2}P[G_2, [G, V_{\text{d}}]]P. \quad (6.1)$$

A third homological generator that removes the residual order-three off-diagonal block changes no order-four low block because its commutator with K is off diagonal. Exact block multiplication gives the complete coefficient

$$\boxed{\Theta_{\text{seq}}^{(4)} = \frac{1}{2}\{F, B\} - S^*RS.} \quad (6.2)$$

For the scalar fixture $\Gamma = 2$, $T = 1/10$, $A = 1/3$, and $C = 5/3$,

$$B = \frac{1}{200}, \quad F = \frac{1}{400}, \quad S = -\frac{1}{15}, \quad \Theta_{\text{seq}}^{(4)} = -\frac{1591}{720000}. \quad (6.3)$$

The exact raw oracle is -1591/720000. The sequential gauge is not generally the direct-rotation standard-SW gauge. Their order-three relative low-space unitary has generator

$$K_P = \frac{1}{2}P[G_2, G]P, \quad \boxed{\Theta_{\text{std}}^{(4)} = \Theta_{\text{seq}}^{(4)} - [K_P, A].} \quad (6.4)$$

For the retained noncommutative fixture, exact arithmetic gives

$$B = \begin{pmatrix} 23/1000 & 2/125 \\ 2/125 & 19/750 \end{pmatrix}, \quad F = \begin{pmatrix} 61/10000 & 31/5000 \\ 31/5000 & 259/22500 \end{pmatrix}, \quad (6.5)$$

$$S^*RS = \begin{pmatrix} 37247/1350000 & -176/84375 \\ -176/84375 & 15887/337500 \end{pmatrix}, \quad (6.6)$$

$$\Theta_{\text{seq}}^{(4)} = \begin{pmatrix} -1476947/54000000 & 128339/54000000 \\ 128339/54000000 & -52517/1125000 \end{pmatrix}, \quad (6.7)$$

$$K_P = \begin{pmatrix} 0 & -857/135000 \\ 857/135000 & 0 \end{pmatrix}, \quad \Theta_{\text{std}}^{(4)} = \begin{pmatrix} -35249/18000000 & -557261/54000000 \\ -557261/54000000 & -243251/3375000 \end{pmatrix}. \quad (6.8)$$

Thus the gauges visibly differ when A is nonscalar. The independent direct-rotation recursion with the off-diagonal inverse Liouvillian reproduces (6.8) exactly. When A is scalar, $[K_P, A] = 0$ and they coincide.

7 Direct additivity and disconnected fixture

Sequential linkedness is not inferred from a factorization of global homological generators. Let X, Y be disjoint components. The operators B, F split into one-component pieces, while the full source is

$$S = S_X \otimes P_Y + P_X \otimes S_Y - (T_X \otimes D_Y + D_X \otimes T_Y). \quad (7.1)$$

The single-excitation terms from DA cancel the corresponding CD terms. On the double-excitation sector,

$$S_{XY} = -(T_X \otimes D_Y + D_X \otimes T_Y) = -(K_X + K_Y)(D_X \otimes D_Y), \quad (7.2)$$

$$R_{XY}S_{XY} = -D_X \otimes D_Y, \quad (7.3)$$

$$S_{XY}^* R_{XY} S_{XY} = B_X \otimes F_Y + F_X \otimes B_Y. \quad (7.3)$$

Equation (7.3) is exactly the mixed part of $(1/2)\{F, B\}$. The orthogonal single-excitation sectors leave only $\Theta_X^{(4)} \otimes P_Y + P_X \otimes \Theta_Y^{(4)}$. Thus each **disconnected multivariate coefficient** at fourth order cancels after its required permutation aggregate. No assertion says every individual ordered term vanishes, and no global-generator factorization is asserted. The triangle inequality over the remaining connected ordered contributions gives the safe $2880z^4$ factor in (5.11).

For a direct product fixture, take $\Gamma_X = 2, T_X = 1/5$ and $\Gamma_Y = 3, T_Y = 1/7$, with both diagonal perturbation blocks zero. The mixed BF contribution is $1/8820$; the mixed S^*RS contribution is also $1/8820$ and cancels it. Therefore

$$\Theta_{X \sqcup Y}^{(4)} = \frac{(1/5)^4}{2^3} + \frac{(1/7)^4}{3^3} = \frac{69827}{324135000}. \quad (7.4)$$

The raw oracle is $69827/324135000$. This fixture executes the direct resolvent identities; the fixed-order literature theorem independently gives connected support for the standard coefficient.

8 Fixed-order spectral-Ritz removal

Let Π_M be nested parity-preserving onsite spectral Ritz projections containing P , commuting with K , and converging strongly to one. Each local $E_e = R^{1/2}T_e$ has finite rank because its low domain is finite rank. Consequently

$$\Pi_M E_e \longrightarrow E_e \quad \text{in operator norm.} \quad (8.1)$$

The compressions $\Pi_M W_f \Pi_M$ are uniformly bounded and converge strongly. Strong convergence becomes norm convergence only after restriction to the finitely many finite-rank ranges entering one fixed connected coefficient. This proves operator-norm convergence of every fixed connected third- and fourth-order coefficient. The finite rooted counts then give the corresponding QPS-norm Ritz removal uniformly in spatial volume.

No global operator-norm convergence of W_f is claimed. The argument is special to orders three and four; it supplies neither a fifth-order modulus nor summability over all orders.

9 Two exact failures of automatic promotion

9.1 A zero low–high tail does not control a moving insertion family

Let

$$\mathcal{H} = \mathbb{C}p \oplus \ell^2\{e_j : j \geq 1\}, \quad K = R = Q, \quad T = |e_1\rangle\langle p|, \quad (9.1)$$

and let Π_M project onto $p, e_1, \dots, e_M$. For every $M \geq 1$,

$$\tau_M = \|(1 - \Pi_M)T\| = 0. \quad (9.2)$$

For $j \geq 2$, define the uniformly bounded positive family

$$C_j = |e_1 + e_j\rangle\langle e_1 + e_j|, \quad \|C_j\| = 2. \quad (9.3)$$

Choosing $j = M + 1$ gives

$$\|(1 - \Pi_M)C_j\Pi_M T\| = 1, \quad (9.4)$$

$$\|(C_j T)^*(C_j T) - (\Pi_M C_j \Pi_M T)^*(\Pi_M C_j \Pi_M T)\| = 1. \quad (9.5)$$

Every fixed C_j becomes exact once $M \geq j$, but the supremum over the moving family fails. Thus even an exactly zero low–high tail does not automatically control a uniform high–high insertion cutoff. The actual Q3 route instead uses (5.5) and finite-rank ranges.

9.2 Orbit-smearing labels do not automatically define a spatial net

On two qubits take $H = X \otimes X$ and $\alpha_t = \text{Ad}(e^{itH})$. Formally disjoint seeds evolve as

$$\begin{aligned} \alpha_t(ZI) &= \cos(2t)ZI + \sin(2t)YX, \\ \alpha_t(IZ) &= \cos(2t)IZ + \sin(2t)XY. \end{aligned} \quad (9.6)$$

Smear the first with $f_+(t) = e^{-t}\mathbf{1}_{t \geq 0}$ and the second with $f_-(t) = e^t\mathbf{1}_{t \leq 0}$. Both have L^1 norm one, and

$$\int f_{\pm}(t) \cos(2t)dt = \frac{1}{5}, \quad \int f_+(t) \sin(2t)dt = \frac{2}{5}, \quad \int f_-(t) \sin(2t)dt = -\frac{2}{5}. \quad (9.7)$$

Thus

$$A_+ = \frac{1}{5}ZI + \frac{2}{5}YX, \quad B_- = \frac{1}{5}IZ - \frac{2}{5}XY, \quad (9.8)$$

but

$$[A_+, B_-] = -\frac{8i}{25}Y \otimes Y, \quad \|[A_+, B_-]\| = \frac{8}{25}. \quad (9.9)$$

The raw norm oracle is $8/25$. Temporal orbit smearing does not automatically turn formal seed support into commuting spatial local algebras. This is not a no-go for every orbit-smear C-star carrier, for a construction with propagation estimates, or for every possible local net.

10 Devil's-advocate review

1. **DISMISSED – the onsite physical splitting was omitted:** that defect is exactly why EXP-000827 corrects EXP-000826. Equation (3.6) restores $\delta_1 \sum_x P_{1,x}$ at the physical zero-source point.
2. **DISMISSED – local rank two violates the linked-cluster hypotheses:** the finite-dimensional construction permits a finite-rank onsite ground projector, and the fixed Ritz gap separates the next level.
3. **UPHELD as an overclaim – a coefficient theorem proves all-order convergence:** the linked theorem is separately fixed in n . It assumes and proves no convergence of the complete series.
4. **UPHELD as an overclaim – the energy estimate holds at physical lambda=1:** it requires $|\lambda|J_M$ below a threshold depending on fixed n, M . No physical smallness check is supplied.
5. **DISMISSED – bare diagonal constants can be reused:** equations (5.2)–(5.5) recompute the augmented low and high blocks, select the orthogonal direct-sum maximum, and retain only the genuinely unchanged cross leg.
6. **UPHELD as a false shortcut – strong Ritz convergence is global norm convergence of W_f :** only its action on finitely many finite-rank ranges is promoted to norm convergence.
7. **DISMISSED – (6.2) is a selected diagram:** (6.1) contains the complete projected order-four terms after the two sequential rotations, and exact block multiplication gives (6.2).
8. **UPHELD – sequential and standard gauges have equal nonscalar matrices:** (6.4) is their exact low-space unitary crosswalk, and (6.7)–(6.8) display the difference.
9. **DISMISSED – every disconnected ordered term vanishes:** that is not claimed. Equations (7.1)–(7.3) cancel each disconnected multivariate permutation aggregate, sufficient for direct additivity and the connected triangle bound.
10. **DISMISSED – $\tau_M = 0$ controls every bounded insertion family:** the moving family (9.3) has insertion tail and Gram difference one.
11. **UPHELD as an overclaim – the Pauli fixture rejects all orbit-smear carriers:** it rejects only automatic spatial commutativity inferred from seed labels without a propagation/locality theorem.
12. **UPHELD as an overclaim – orders three and four close the oscillator parent:** no fifth-order envelope, order summability, phasewise intertwiner, or broken-sector GNS coercivity estimate is supplied.
13. **VALID with mitigation – passing verification closes physical parents:** the engines bind formulas, authorities, hashes, lifecycle, and scope only. Passing changes no claim tier and closes none of the five open parents.

Independent adversarial review is invited especially on the zero-source onsite allocation, finite-rank ground projector hypothesis, fixed-order versus all-order quantifiers, physical- $\lambda = 1$ smallness boundary, the orthogonal maximum in (5.5), all seven exact fractions, fourth-order signs and factors, the standard/sequential gauge correction, disconnected resolvent cancellation, finite-rank Ritz restriction, both counterfixtures, and every scope firewall.

11 Reproduction and exact verification pins

The R-167 v2.6 primary, non-importing independent, and integrated scripts are

1. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py`;
2. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py`;
3. `codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py`.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
    'codes\foundations\pre_a_cp1_st8_q3lock_' +
    'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py') --no-store
& $py -X utf8 ($r167 + '_independent.py') --no-store
& $py -X utf8 ($r167 + '_verify.py') --no-store
```

The expected authority-complete totals are primary PASS 442/442, non-importing independent PASS 246/246, and integrated PASS 425/425. The exact raw script SHA-256 pins are

```
R-167 primary:
f0f177b3b7c03f4e02795499cc5d5a8e432388269d51ab1d323a85c3974a6077
R-167 independent:
a54eb34db7944a2423963409cea7205ec362a31ed0259e23ba34c303b4213253
R-167 integrated:
4d075cad0122cdf94b2c422fba361af9a00626050808936c8848c8fb6ac06399
```

The independent pin is its raw script hash, not a canonical stdout payload hash. The primary symbolic and non-importing standard-library rational engines independently recompute every new matrix and fraction. The integrated engine binds them to both exploration records, events 617/618, R-167 v2.6, all four closed and five open gates, both negative authorities, generated surfaces, stored runs, and checkpoint lifecycle. Any missing check invalidates the total.

Source-form validation uses

```
$note = (
    'claims\C6-SPACETIME-SIGNATURE\notes\' +
    'pre-a-q3lock-zero-source-fixed-order-fourth-linked-' +
    'and-orbit-locality-boundary-checkpoint-260812-v1.5.tex.txt'
)
& $py -X utf8 verification\scripts\build_note_pdf.py '
--no-compile $note
```

The source-form command creates no PDF. No per-lemma or intermediate PDF is issued. This source is the one R-167-only v2.6 gate-level checkpoint source; its paired PDF may be issued only after the primary, non-importing independent, integrated, formal-authority, generated-surface, source-form, freshness, dual-extraction, strict-release, and visual-review gates pass. Before issuance the lifecycle is exactly the three-field DEFERRED record. After issuance it must have exactly the eight fields **status**, **source**, **pdf**, **source_sha256**, **pdf_sha256**, **pages**, **workflow**, and **visual_qa**.

R-168 v1.3 remains historical and is not reissued; no R-168 count or hash is introduced by this checkpoint.

12 Exact no-overclaim and dependency boundary

The connected rank-two oscillator-elimination parent is **OPEN**. Fixed-order linked support, exact third/fourth coefficients, and fixed-order Ritz removal give no fifth-order coefficient envelope, no n -summability, no convergent global or local SW series, no boundary/source-uniform unbounded elimination, and no physical $\lambda=1$ theorem.

The broader infinite-dimensional rank-two and broken-sector spectral parents are **OPEN**. Nothing here gives an exact oscillator two-phase QPS, phasewise intertwiner, temporal-mass estimate, oscillator-lattice GNS coercivity, broken-sector GNS gap, or physical mass gap.

The common-alpha parent is **OPEN**. The orbit-smear counterfixture rejects automatic spatial commutativity from formal seed labels; it does not rule out another carrier, an unsplit construction, state-weighted convergence, a propagation-controlled orbit-smear algebra, or common-alpha existence. There is no all-shape/all-exhaustion Q3 Cauchy estimate, identified local generator, or KMS quotient.

The Round-1 parent is **OPEN**. The result changes neither the evidence role freeze nor the prospective holdout state. It does not promote internal T0 evidence to a physical claim.

Nothing here proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, a below-empty free-energy sign, physical vacuum selection, emergent time, a universal propagation cone, gravity, C6 emergence, CP1, physical Sector A, or Pre-A closure. The C6-SPACETIME-SIGNATURE claim card and tier are unchanged.

13 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v2.6.
Active exploration:	EXP-000826 as corrected by EXP-000827.
Event provenance:	Event 617 remains historical; event 618 supersedes its source qualifier and two QPS constants only.
Precise statement:	At zero source and each fixed M, every separately fixed standard-SW coefficient is linked; for fixed (n,M) and sufficiently small coupling the fixed-order ground-energy error is volume extensive; actual Q3 orders three/four have the stated QPS/Ritz bounds; and fixed finite (Lambda,M) has the complete sequential fourth-order coefficient and standard-SW gauge crosswalk.
Scope:	T0 additive analytic/exact R-167 v2.6 at zero source and fixed order/cutoff only; no load-bearing parent, claim tier, Sector A, C6, CP1, or Pre-A closure.
Dependencies:	R-167 v2.5; EXP-000826 as corrected by EXP-000827; Bravyi-DiVincenzo-Loss fixed-order linked-cluster theorem; zero-source incident onsite allocation and augmented-edge low/high block estimates.
Precise positive scope:	Zero-source fixed-M standard-SW connected coefficients at every separately fixed order n; fixed-(n,M) small-coupling volume-extensive energy accuracy; actual zero-source Q3 third/fourth QPS and Ritz removal; complete fixed-(Lambda,M) sequential fourth order and standard-SW gauge crosswalk.
Closed children:	Exactly the four IDs in Section 2, in their corrected scopes.
Negative results:	Exactly the two automatic-implication failures in Section 2.
Open parents:	Exactly the five load-bearing IDs in Section 2 remain OPEN.
Core fourth order:	$\Theta_{4,seq} = (1/2)\{F,B\} - S*RS;$

Theta4_std=Theta4_seq-[K_P,A].

Exact scalar oracle: -1591/720000.

Disconnected oracle: Equal mixed 1/8820 terms cancel; local sum is 69827/324135000.

Orbit-smear oracle: Commutator norm 8/25.

Corrected a_tilde: 32063/500000.

Corrected rho_tilde: 2918597/30000000.

Corrected w_tilde: 584161/6000000.

Corrected QPS3: 1411418996889/381469726562500.

Corrected QPS4: 2314877826589220601/29802322387695312500.

Evidence grade: ANALYTIC + EXACT + primary + non-importing independent + integrated; R-167-only checkpoint.

Reproduction command: Run the three final verifier commands in Section 11 from repository root, each with --no-store.

Expected output: PASS 442/442; PASS 246/246; PASS 425/425.

Primary raw SHA-256: f0f177b3b7c03f4e02795499cc5d5a8e432388269d51ab1d323a85c3974a6077.

Independent raw SHA-256: a54eb34db7944a2423963409cea7205ec362a31ed0259e23ba34c303b4213253.

Integrated raw SHA-256: 4d075cad0122cdf94b2c422fba361af9a00626050808936c8848c8fb6ac06399.

Falsification gate: Any zero-source allocation, fixed-order quantifier, small-coupling, augmented block, exact fraction, fourth-order sign/factor, gauge, additivity, Ritz, counterfixture, assertion-count, hash, provenance, or scope mismatch.

Tier before / after: C6 T1 / C6 T1; no claim or tier change.

No-overclaim statement: No n-to-infinity or all-order convergence, physical lambda=1 smallness, fifth-order or unbounded oscillator elimination, phase transfer, actual Q3 common alpha, generator/KMS quotient, oscillator GNS gap, Round-1, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.

Next required action: Prove a remainder-controlled or convergent all-order weighted recursion and a viable spatially local carrier before any oscillator phase/GNS or common-alpha promotion.

PDF policy: No per-lemma or intermediate PDF. One R-167-only v2.6 gate-level pair is issued only after all proof, authority, extraction, release, and visual gates.

R-168 firewall: R-168 v1.3 remains historical and is not reissued; no R-168 verification count or hash is part of this document.